

Answers to Practice Papers — 4

1. (c) %
2. (b) 6
3. (b) 255 bytes
4. (b) Optical fiber
5. (b) Bridge
6. (c) append()
7. (a) Mult()
8. (a) Primary key
9. (c) error
10. (c) 2ndName
11. (a) 'r', 'r+', 'w', 'w+', 'a', 'a+'
12. (a) Built-in functions, functions defined in a module, and user-defined functions
13. (b) file.write()
14. (a) [1, 2, 3]
15. (b) 'rb+'
16. (b) The last element added to the stack
17. (a) Both (A) and (R) true and (R) is the correct explanation for (A).
18. (b) Both (A) and (R) true and (R) is not the correct explanation for (A).
19.

```
Val = int(input("Value:"))          # Error 1
Adder = 0
for C in range(1,Val,3):             # Error 2
    Adder+=C
    if C%2==0:                       # Error 3
        print (C*10)
    else:                             # Error 4
        print (C)
print(Adder)
```

20. Advantages of Star Topology over Bus Topology

- (a) Faster communication as compared to Bus topology
- (b) Independent line of connection allows freedom of removing or adding nodes from the network

Disadvantages of Star Topology over Bus Topology

- (a) Expensive as compared to Bus topology
- (b) Long cable length

OR

Web Browser: A web browser is a software application for accessing information on the World Wide-Web. When a user requests a web page from a particular website, the web browser retrieves the necessary content from a web server and then displays the page on the user's device. **Example:** Google Chrome, Safari, Edge, Opera

21. **Syntax Error:** An error of language resulting from code that does not confirm to the syntax of the programming language.

Example:

```
a = 0
while a < 10 # : is missing as per syntax
```

```
a = a + 1
print ( a)
```

Runtime Error: A runtime error is an error that causes abnormal termination of program during running time.

Example:

```
A=10
B= int(input("Value:"))
print (A/(B))
# If B entered by user is 0, it will be run-time error
```

22. A primary key is a unique identifier for each record in a database table. It is used to enforce the integrity of the data and ensure that each record can be uniquely identified. For example, consider a table "customer" that stores information about customers in a retail store. A primary key for this table could be the "customer_id" column, which assigns a unique ID number to each customer in the table. This ID is used to refer to the customer in other tables and ensure that there are no duplicate records for a single customer in the "customers" table.
23. (a) FTP – File Transfer Protocol and NIU–Network Interface Unit
 (b) An Ethernet card is a hardware component that connects a computer to a network, allowing it to send and receive data using the Ethernet protocol. It is available in various form factors, including PCI, USB, and wireless.
24. (a) 1 20 P\$
 4 30 P\$R\$
 9 60 P\$R\$S\$
 (b) (ii) 20#40#45# and (iii) 50#20#40#
 Max value for BEGIN 3
 Max value for LAST 4
25. Constraints in MySQL are used to enforce rules on data stored in a database table and ensure the integrity of the data. The purpose of constraints is to prevent incorrect data from being stored in the table and to ensure that the data stored in the table meets certain conditions, such as unique values, minimum and maximum values, and relationships between tables. Constraints can be used to enforce rules on individual columns or across multiple columns in a table. They are an important part of database design and play a crucial role in maintaining the consistency and accuracy of data in a database.

OR

Aggregate functions in MySQL are used to perform calculations on a set of values and return a single value. These functions include:

SUM: Returns the sum of all values in a specified column.

AVG: Returns the average value of all values in a specified column.

COUNT: Returns the number of rows in a table, or the number of non-NULL values in a specified column.

MIN: Returns the minimum value in a specified column.

MAX: Returns the maximum value in a specified column.

26. (a)

StudentID	Name	Age	Course	StartDate
1	John	23	Math	2020-01-01
1	John	23	Science	2020-02-01
2	Jane	25	History	2020-03-01
3	Alex	22	English	2020-04-01

(b) (i)

departure_city
Delhi
Mumbai
Kolkata
Bengaluru

(ii)

count(*)
5

(iii)

departure_city
Bengaluru
Delhi

(iv)

departure_city	arrival_city
Kolkata	Bengaluru
Delhi	Chennai
Bengaluru	Delhi
Mumbai	Kolkata
Delhi	Mumbai

```
27. defdisplay1():
    c=0
    file=open('INDIa.TXT','r')
    c=0
    for LINE in file:
        Words = LINE.split()
        for W in Words:
            if W=="India":
                c=c+1
    print (c)
    file.close()
```

OR

```
def countword():
    f=open("quotes.txt","r")
    count=0
    new=""
    x=f.read()
    word=x.split()
    for x1 in word:
        new += x1
        if len(x1)==4:
            count=count+1
    print("Total words ",count)
    f.close()
```

28. (a) Cartesian Product

Degree = 5

Cardinality = 9

(b) Count(*)

3

3

3

29. def ZeroEnding (SCORES) :

SZero = 0

for i in SCORES :

if i % 10 == 0:

SZero += i

print ("sum of numbers ending with zero : ", SZero)

SCORES=[200,456,300,100,234,678]

ZeroEnding (SCORES)

30. def Pop(Stak) :

if isEmpty(Stak):

print("Underflow")

return -1

else:

Ele = Stak.pop()

if len(Stak)==0:

top=None

else :

top=len(Stak)-1

return Ele

OR

N=[12, 13, 34, 56, 21, 79, 98, 22, 35, 38]

def PUSHEI(S,N):

S.append(N)

def POPEI(S):

if S!=1I:

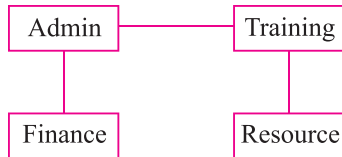
return S.pop()

```

else:
    return None
    ST=[]
    for k in N:
        if k%4==0:
            PUSH(ST,k)
while True:
    if ST!=II:
        print(POP(ST).end=" ")
    else: break

```

31. Server Connectivity



- (i) Layout
- (ii) Admin
- (iii) SWITCH/HUB
- (iv) ADMIN & FINANCE
- (v) Optical Fiber

32. (a) 1350#8#

- (b) Statement 1 - mysql.connector.connect
- Statement 2 - conn.cursor()
- Statement 3- cursor.fetchone()

OR

- (a) coMpUtER****
- (b) Statement 1 - cno, companyname, model, price
- Statement 2 - cursor = conn.cursor()
- Statement 3 - cursor.execute(sql)

33. import pickle

```

def update_product_stock(prod_code, new_stock):
    try:
        with open("PRODUCT.DAT", "rb") as f:
            product = pickle.load(f)

        # Check if the prod_code matches the record in the file
        if product["prod_code"] == prod_code:
            product["stock"] = new_stock
            with open("PRODUCT.DAT", "wb") as f:
                pickle.dump(product, f)
            print(f"Stock for product code {prod_code} has been updated to {new_stock}")
        else:
            print(f"No record found for product code {prod_code}")
    except EOFError:
        print("No records found in the file")

```

OR

```
import pickle

def find_max_salary(file_path):
    max_salary = 0
    max_salary_member = None

    with open(file_path, 'rb') as file:
        while True:
            try:
                member = pickle.load(file)
                memno, mname, salary = member
                if salary > max_salary:
                    max_salary = salary
                    max_salary_member = member
            except EOFError:
                break

    if max_salary_member is not None:
        memno, mname, salary = max_salary_member
        print("Max Salary:", salary)
        print("Member No:", memno)
        print("Member Name:", mname)
    else:
        print("No member found in the file.")

find_max_salary("Member.dat")
```

34. (i) Select * From Transact Where Type = ' Deposit ' ;
(ii) Select Ano,Amount From Transact
Where Dot >= ' 2017-10-01 ' And Dot <= ' 2017-10-31 ' ;
(iii) Select Account.ano,Aname,Dot From Account,Transact
Whereaccount.ano=Transact.ano And Amount <=3000;

OR

(iii) ANO	ANAME
103	Ali Reza
105	Simran Kaur

35. (a) F = open("STUDENT.DAT",'wb')
(b) pickle.dump(L,f)
(c) r = pickle.load(f)
(d) (i) 'r+' opens a file for both reading and writing. File object points to its beginning.